

Chi coupling channel activity in TmFeO3 Gamma_2 phase

Left: ground-state lambda generation | Right: H//z detectability (M_z^Tm)

Ground state: H[λ²] pattern generated by each channel

$M_z^{Tm} = \eta^z_{Tm} \cdot \lambda^2 = \sigma_C \cdot \lambda^2$ (H//z detector coupling)

Channel	η^{α}_{Fe}	Fe spin pattern	$\eta \cdot S$ product	λ^2 sigma pattern	Active?
chi2z	$\sigma_G = (+-+-)$	$G_z \cdot \sigma_G$	$\sigma_G \cdot \sigma_G = \sigma_F$	σ_F (uniform)	YES ★
chi2x	$\sigma_A = (++--)$	$F_x \cdot \sigma_F$	$\sigma_A \cdot \sigma_F = 0$	none ($F_x \approx 0$)	no
chi2y	$\sigma_C = (+--+)$	$S^y = 0$	—	none	no
chi5z	$\sigma_G = (+-+-)$	(any)	cancels	none	no ✗
chi5x	$\sigma_A = (++--)$	(any)	cancels	none	no ✗
chi5y	$\sigma_C = (+--+)$	(any)	cancels	none	no ✗
chi7z	$\sigma_G = (+-+-)$	(any)	cancels	none	no ✗
chi7x	$\sigma_A = (++--)$	(any)	cancels	none	no ✗
chi7y	$\sigma_C = (+--+)$	(any)	cancels	none	no ✗

Channel	λ^2 sigma pattern	$\sigma_C \cdot \lambda^2_{sigma}$	M_z^{Tm}	Detectable by H//z?
chi2z	$\sigma_F = (++++)$	$\sigma_C \cdot \sigma_F = 0$	$= 0$	NO ✗
chi2x	σ_A (tiny)	$\sigma_C \cdot \sigma_A = 0$	$= 0$	NO ✗
chi2y	dead	—	0	NO ✗
chi5*	dead	—	0	NO ✗
chi7*	dead	—	0	NO ✗
—	need σ_C	$\sigma_C \cdot \sigma_C = 4$	$\neq 0$	YES

★ chi5/chi7 cancel because even/odd bonds share the same Fe sublattice within each orbit pair: sign_57=(+1)+(-1)=0 identically.
chi2y inactive because $S^y_{Fe}=0$ in Γ_z by symmetry.
chi2x inactive because $\sigma_A \cdot \sigma_F=0$ (small F_x canting is σ_F , not σ_A).

H//z couples to $M_z^{Tm} = \sum_j \eta^z_{Tm,j} \cdot J^z_{Tm,j}(\text{local})$
 $J^z_{Tm,j}(\text{local}) = c_2 \cdot \lambda^2_j$ (leading-order matrix element)
 $\eta^z_{Tm} = \sigma_C = (+,-,-,+)$ [from Tm frame diagonals R_{zz}]

For any chi channel, λ^2 must carry σ_C pattern to be detected by H//z.
No chi2 channel produces $\sigma_C \lambda^2$ from Gamma_2 background:
chi2z $\rightarrow \sigma_F$, chi2x $\rightarrow 0$, chi2y $\rightarrow$ dead.

CONCLUSION: The (omega_qAFM, omega_E12) cross-peak is FORBIDDEN under H//z for ALL chi channels in the Gamma_2 phase.